Unemployment Insurance Extensions, Labor Market Concentration, and Match Quality

by

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Abstract

I investigate whether the effects of UI extensions are different for workers exposed to higher levels of local labor market concentration, a potential source of employer market power. I exploit measurement error in state unemployment rates that led to quasi-random assignment of UI durations in the U.S. during the Great Recession. Using matched employer-employee data from the Longitudinal Employer-Household Dynamics program, I find that UI extensions lengthen nonemployment durations by one week and cause economically meaningful but not statistically significant increases in earnings. The UI-earnings effect is significantly lower at higher levels of concentration, while there is no difference in the UI-duration effect. The lower UI-earnings effect is driven by the extremes of the distribution of concentration. My results suggest that match improvements from UI are attenuated at higher levels of concentration.

Keyword: Unemployment insurance, labor market concentration, local labor markets, earnings, nonemployment duration

JEL Classification: J31, J42, J65

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1 Introduction

Unemployment insurance (UI) policy in the U.S. is set at the state and federal level, but labor markets are local. When studying the effect of UI, and job search outcomes in general, it is important to understand the role of local labor market conditions. For example, Kroft et al. (2013) show that the effect of a worker's nonemployment duration on callback rates varies substantially with local labor market tightness. Other studies also have found that the effect of UI along different margins depends on labor market conditions (Kroft and Notowidigdo, 2016; Schmieder et al., 2012; East and Kuka, 2015). Another dimension along which local conditions influence labor demand and supply is local labor market concentration. Concentration is associated with lower posted pay, wages, earnings, and total compensation (e.g., Azar et al., 2022; Marinescu et al., 2021; Rinz, 2022; Qiu and Sojourner, 2019). Higher levels of concentration also are associated with lower flows out of nonemployment and worse outcomes after a nonemployment spell (Berger et al., 2022a; Dodini et al., 2020). Despite extensive literatures documenting the separate importance of both UI and concentration for workers' outcomes, no prior work studies their interaction.

I fill this gap in the literature by studying how the effect of UI benefit extensions on nonemployment duration and subsequent earnings varies with local labor market concentration. I use data on 25 million nonemployment spells from 19 million workers spread across 28 states and an empirical strategy that exploits measurement error in state unemployment rates that led to quasi-random assignment of potential UI durations in the U.S. during the Great Recession.

Ex-ante, it is not clear whether the effects of UI extensions would be different in more vs. less concentrated labor markets as no prior work has addressed this question. On the one hand, longer UI benefits provide an additional outside option that allows nonemployed workers to search more selectively and potentially improves their bargaining power (e.g. Acemoglu and Shimer, 1999). This implies that workers exposed to higher levels of concentration, a potential source of monopsony power, would benefit differentially from such a change to their bargaining power.[1] On the other hand, workers in more concentrated markets might be limited in their selectivity because fewer firms directly compete for them. And at baseline, labor markets with higher levels of concentration have lower job-finding rates (Marinescu et al., 2021; Berger et al., 2022a). If the UI-duration elasticity is

[1] Labor market concentration is a potential source of employer market power (Berry et al., 2019; Syverson, 2019; Eeckhout, 2021).

the same at different levels of concentration, then the benefits of selectivity could be offset by longer nonemployment durations in more concentrated markets due to baseline differences in job-finding rates and negative duration dependence.

My empirical strategy allows me to overcome the biggest challenge to identifying the effect of UI extensions: benefits become more generous when labor markets worsen. In response to the Great Recession in 2008, Congress extended the duration of UI benefits in states with higher unemployment rates. To avoid the endogeneity problem of longer benefits in states with worse labor market conditions, I exploit measurement error in the state-level unemployment rates used to assign maximum potential benefit durations. This approach was first used by Chodorow-Reich et al. (2019) to study the contribution of UI extensions to the aggregate unemployment rate.[2] I use as the treatment variable the difference in actual UI benefit duration as assigned using the real-time unemployment rate, which is partly estimated using survey data and subject to measurement error, and the counterfactual duration as assigned using subsequent revisions to the unemployment rate. The resulting "UI error" is the amount of UI benefit duration attributable to measurement error and not the underlying labor market conditions.

This measurement error approach allows me to identify the effect of UI extensions using weaker identification assumptions than prior work in the U.S. context. Most studies of UI extensions (as opposed to changes in benefit levels) in the U.S. rely on variation in potential benefit duration across states and time (e.g. Rothstein, 2011; Farber and Valletta, 2015; Farooq et al., 2020).[3] This variation requires the assumption that the controls fully account for differences in underlying labor market conditions across treated and untreated states. It is difficult to establish the validity of this assumption because differences in labor market conditions generate the variation in benefits (Schmieder and von Wachter, 2016). A small set of papers avoids this endogeneity issue by using either politically-motivated changes to UI generosity or kinks in benefit schedules, but they are limited to one or two states and so their findings do not necessarily generalize (Card and Levine, 2000; Landais, 2015; Johnston and Mas, 2018). Instead, I overcome the endogeneity of benefit extensions by isolating changes in UI durations due to plausibly exogenous measurement error

[2]Chodorow-Reich et al. (2019) estimate that UI extensions had a very small effect on the unemployment rate during the Great Recession, increasing it by at most 0.3 percentage points.

[3]Schmieder and von Wachter (2016) review the literature on UI benefit levels. The same identification issues in the U.S. exist for these studies, with regression kink designs sometimes possible (Landais, 2015; Card et al., 2015).

using a much larger sample.

I first estimate the overall impact of UI extensions, without accounting for concentration. My preferred approach includes fixed effects for the start of a nonemployment spell, the worker's state of employment, and local labor market of employment prior to the spell. I define local labor markets (LLMs) as the combination of a commuting zone and industry, so the LLM fixed effects are, in practice, commuting zone-by-industry fixed effects.[4] This fixed effects structure isolates variation in UI duration occurring through the UI errors that is orthogonal to systematic differences in labor market conditions over time and across places. I use quarterly matched employer-employee data from the Longitudinal Employer-Household Dynamics (LEHD) program at the U.S. Census Bureau over the period 2004-2014. To focus on those likely to be UI-eligible at the time of separation, I restrict my attention to prime-age workers (aged 25-54) who were employed for at least two consecutive quarters prior to the start of a nonemployment spell.

I find that extending UI benefits by one quarter (13 weeks) causes nonemployment durations, as measured by the number of full quarters with zero earnings, to be longer by 0.8 weeks. This estimate is statistically significant at the 5% level, demonstrating that my approach and sample allow me to identify small effects. The point estimate represents a 2.8% increase relative to the mean nonemployment duration. This effect size is in line with other recent studies (Rothstein, 2011; Farber and Valletta, 2015; Farber et al., 2015; Kroft and Notowidigdo, 2016), though it is smaller than the effects identified off of politically-motivated changes or benefit kinks (Card and Levine, 2000; Johnston and Mas, 2018; Landais, 2015).

In terms of earnings, my preferred measure of match quality, I estimate that extending UI benefits by one quarter causes earnings at re-employment to increase by 0.9%, which is not statistically significant. I obtain very similar estimates of the duration and earnings effects using a slightly different specification that uses state-time variation in potential UI benefit duration while controlling flexibly for the concurrent revised unemployment rate. This model could not be estimated in the absence of measurement error because the UI duration would be perfectly collinear with the unemployment rate. The similarity of the estimates from my preferred approach and this alternative model confirms that the underlying identifying variation in my estimates is due to measurement

[4] I do not observe occupation information. Local labor markets based on occupations also exhibit substantial variation over time and across places (Handwerker and Dey, 2022).

error and not labor market conditions. Farooq et al. (2020) study UI extensions during the Great Recession using state-time variation in benefits and the same data as this paper. When examining a comparable time period to this paper (2008-2013), they find that a one quarter UI extension causes quarterly earnings to increase by 0.5%, similar to my preferred estimate of 0.9%. Johnston and Mas (2018) find a small but not significant increase in re-employment earnings following a benefit cut in Missouri.

Outside the U.S., most studies estimate negative UI-wage effects (e.g., Card et al., 2007; Lalive, 2007; Schmieder et al., 2016), though Nekoei and Weber (2017) find a positive 0.5% effect following a nine-week extension in Austria. These papers exploit discontinuities in UI policy, making it difficult to interpret their estimates for workers away from the discontinuities. In addition,, it is not clear how well these estimates would translate to the U.S. context where wage determination is more decentralized.

I next estimate how the effect of UI extensions varies across the distribution of local labor market concentration. I interact the UI error with HHI in the quarter in which the worker separates from employment (i.e., a triple difference model). Importantly, this approach does not require labor market concentration to be exogenous. Identification still stems from exogenous variation in potential UI duration, except now I focus on how local labor market concentration moderates the effect of UI on workers. This requires the additional assumption that the effect of UI extensions in more concentrated markets is an accurate counterfactual for their effect in less concentrated markets. To measure concentration, I follow Berger et al. (2022b) and construct a quarterly payroll version of the HHI, which is the sum of squared payroll market shares across firms in the same local labor market, using data from the Longitudinal Business Database (LBD) from the U.S. Census Bureau. The LBD, unlike the LEHD, contains consistent firm identifiers across states, allowing me to measure concentration in markets that cross state borders.

I find that the effect of UI extensions on the length of nonemployment spells is not statistically significantly different across the distribution of local labor market concentration. For a 0.10 point change in HHI, the difference between a low-concentration market and a high-concentration market according to the U.S. *Horizontal Merger Guidelines*, the effect of a one quarter UI extension on nonemployment duration increases by 0.01 weeks. This difference represents just 0.03% of the mean nonemployment duration. The interaction effect is precisely estimated, as I can rule out increases

in the UI-duration effect larger than 0.16 weeks for a change in HHI of 0.10.

The effect of UI extensions on earnings is less positive at higher levels of concentration. The estimated interaction term is statistically significant at the 5% level and is economically meaningful. For a local labor market with an HHI of 0.15, the UI-earnings effect is 0.96%. At an HHI of 0.25, meanwhile, the UI-earnings effect is about half as large: 0.49%. This UI error-HHI interaction term also is precisely estimated: for a change in the HHI of 0.10, I can rule out interaction effects for earnings outside of the interval ranging from -0.89% to -0.06%. This lower UI-earnings effect in more concentrated markets is driven by effects at the extremes of the HHI distribution: when I use log HHI, the UI-HHI interaction term is smaller in magnitude and no longer significant. The fact that re-employment earnings are relatively lower in more concentrated markets without any difference in the length of nonemployment durations is consistent with local labor market concentration as a source of employer market power.

Are there ways other than the immediate earnings effect that extended UI benefits might impact workers in more concentrated labor markets? To answer this question, I make use of the matched employer-employee data and examine how the mobility of nonemployed workers differs across the distribution of local concentration following a UI extension. The likelihood of switching workplaces, local labor markets, and industries as a result of a UI extension is larger in more concentrated labor markets, but the magnitudes of these interaction effects are not meaningfully large. In addition, longer UI benefits do not induce workers to match into less-concentrated labor markets, regardless of the degree of concentration.

This paper makes two main contributions. First, I introduce the measurement error approach to the literature on UI and worker-level outcomes. This approach allows for weaker identification assumptions than what is typically required in the U.S. context. My estimates of the UI-duration effect are smaller than other estimates from the U.S., likely due to the temporary nature of the extensions I study. I also find a somewhat larger UI-earnings effect, though this estimate is not statistically different from zero.

Second, I provide the first estimates of the interaction of UI and concentration. Most of the concentration literature focuses either on job-stayers (Bassanini et al., 2021) or a mix of job-stayers and job-leavers. In contrast, I focus on nonemployed workers likely seeking employment. A small number of papers examine how concentration relates to job search outcomes. Jarosch et al. (2019)

and Berger et al. (2022a) embed concentration in a search model in which the size of an employer endows them with labor market power. In both cases, higher levels of concentration are associated with worse outcomes for workers due to their limited outside options. Dodini et al. (2020) study the impact of exogenous separations on workers exposed to different levels of concentration and find that workers exposed to higher levels of concentration have lower earnings at re-employment. I go beyond these papers by studying how local labor market concentration interacts with UI, which is independently important for job search outcomes. My results show that the UI-duration effect does not vary with concentration but that the UI-earnings effect is less positive in more concentrated labor markets, especially at the highest levels of concentration, in line with Dodini et al. (2020). Given the existing evidence linking changes in UI generosity with changes in worker bargaining power (Dahl and Knepper, 2022, i.e.,[]), these results imply that the UI extensions I study did not lead to large enough increases in the bargaining power of workers to offset the negative outcomes associated with concentration.

My results show that UI extensions during the Great Recession did not lead to large moral hazard costs by subsidizing unproductive search. Instead, nonemployed workers used the additional benefits to avoid decreases in earnings. These findings align with existing research that indicates the moral hazard cost of UI during downturns is low (e.g. Rothstein, 2011; Kroft and Notowidigdo, 2016). In addition, the findings related to concentration suggest that policies aimed at reducing high levels of local labor market concentration, such as antitrust enforcement, might have complementary benefits in the form of improving the effectiveness of UI.

2 Conceptual Framework

It is theoretically ambiguous how the effect of UI extensions on the quality of new matches would vary with local labor market concentration. In general, UI extensions cause workers to be out of work for longer as a result of reduced search effort (the duration effect) but also may induce workers to be more selective in their search and find a higher paying job (the selectivity effect). The sign and magnitude of the UI-earnings effect is governed by the relative sizes of the elasticity of duration and the elasticity of selectivity, with larger duration effects associated with more negative UI-earnings effects (Schmieder and von Wachter, 2016; Nekoei and Weber, 2017).

Concentration is associated with several features of local labor markets that could influence the duration effect of UI extensions. For example, local labor markets with higher degrees of concentration have fewer new hires (Marinescu et al., 2021), firms in these markets post fewer vacancies, and average flows out of nonemployment are lower compared to less concentrated markets (Berger et al., 2022a) . This evidence implies that the baseline job-finding rate for nonemployed workers is lower in more concentrated markets compared to less concentrated markets.

It is unclear whether or how additional UI benefits would differentially change job-finding rates across the distribution of concentration. If the UI-induced drop in the job-finding rate is constant across the distribution of concentration, then, for the same change in potential UI duration, we should expect the UI-duration effect to be larger in more concentrated markets due to the lower baseline job-finding rate in those markets.[5] This prediction corresponds to a positive UI-HHI interaction effect with respect to nonemployment duration. The same prediction would hold if the UI-induced drop in the job-finding rate is larger in more concentrated labor markets.

There is no clear prediction if workers in more concentrated labor markets reduce their search effort following a UI extension by less than those exposed to lower baseline levels of concentration. In such a scenario, the sign of the UI-concentration interaction effect depends on the size of the difference in baseline job-finding rates and the relative drop in the job-finding rate across the distribution of HHI. Because both the choice of search effort and the degree to which UI extensions change that effort are unobserved, it is not possible to determine how the UI-duration effect changes at different levels of concentration. Adding to this ambiguity, it also is possible that, during recessions, the baseline job-finding rate is constant across the distribution of concentration as the existing evidence does not account for differences across the business cycle.

In terms of earnings, a large and growing literature has shown that concentration is associated with lower posted pay (Azar et al., 2022), wages (Marinescu et al., 2021), earnings (Rinz, 2022; Benmelech et al., 2022), and total compensation (Qiu and Sojourner, 2019), consistent with concentration being a potential source of monopsony power. Many search models, meanwhile, predict that UI extensions can increase worker bargaining power and lead to better match quality (higher earnings) by providing an additional outside option that allows workers to search for longer

[5]In this stylized example, I assume that concentration arises exogenously. In reality, however, concentration is determined in equilibrium. For the purposes of exposition this is a reasonable assumption because UI extensions are unlikely to directly affect concentration.

(e.g. Acemoglu and Shimer, 1999; Chetty, 2008). These models do not account for labor market concentration, but their predictions imply that workers exposed to higher levels of concentration will differentially benefit if the extensions provide a sufficient improvement in their outside option and strengthen their bargaining power relative to employers, reducing or offsetting the negative outcomes associated with concentration.[6]

There is limited direct evidence in the UI literature on how UI policy changes impact the bargaining power of affected workers. Dahl and Knepper (2022) study large UI reforms in several U.S. states that reduced both the level and duration of UI benefits. Using data on establishments located in differently-treated states but within the same firm, they find that starting salaries decrease by 7% and posted salaries for the same job in the same firm decrease by 6% in reform states. These results for posted pay indicate that reductions in UI generosity cause worker bargaining power to fall when the outside option of remaining nonemployed becomes less valuable.[7] The theoretical predictions discussed above, as well as this empirical evidence of UI generosity influencing worker bargaining power, imply that the UI-earnings effect will be more positive in local labor markets with higher levels of concentration.

There also are reasons to expect that the UI-earnings effect will be more negative in markets with higher levels of concentration. Concentration is associated with a variety of negative labor market outcomes, including lower earnings. If UI extensions do not sufficiently increase the bargaining power of workers exposed to higher levels of concentration relative to employers, then, following a change in UI policy and holding everything else constant, we would expect the earnings at reemployment for the average nonemployed worker in a highly concentrated market to be lower than that of the average nonemployed worker in a less concentrated market. This prediction follows not only from the papers discussed above but also static models of labor market concentration (Boal and Ransom, 1997; Arnold, 2020) and dynamic models of labor market power in which concentration

[6] I use the term "outside option," which is relevant for wage bargaining models, loosely. In wage posting models with random search such as Burdett and Mortensen (1998), additional UI benefits enter as an increase in the reservation wage. In directed search models such as Acemoglu and Shimer (1999), more generous UI may allow workers to search in a "higher" submarket.

[7] Jäger et al. (2020) also study UI reforms in Austria that increased the level but not the duration of benefits for some, but not all, nonemployed workers. They find that wages for eligible incumbent workers are insensitive to changes in UI benefit levels, implying that the value of nonemployment as an outside option has little bearing on wages. They further show that this insensitivity persists among groups thought to have limited bargaining power such as women, blue-collar workers, and younger workers. It is reasonable to expect a different effect in the U.S. context, however, where worker bargaining power is lower and wage determination is more decentralized.

is embedded (Jarosch et al., 2019; Berger et al., 2022b,a).

A similar prediction holds if we consider the trade-off faced by nonemployed workers following a UI extension. These workers can become more selective in their job search, which pushes their expected re-employment earnings upward. But the additional benefits also lead to reduced search effort, which pushes their expected earnings downward as a result of negative duration dependence. Nonemployed workers exposed to higher levels of concentration, however, might be limited in their ability to become more selective as, conditional on the size of the local labor market, fewer employers directly compete for them relative to workers exposed to less concentration. These predictions suggest that the UI-earnings effect should be more negative in local labor markets with higher levels of concentration.

Taken together, the arguments in this section illustrate why it is theoretically ambiguous how the effect of UI extensions would vary with local labor market concentration. The question of the sign and magnitude of the interaction between UI extensions and local labor market concentration is therefore an empirical one. This ambiguity holds for both the UI-duration effect and the UI-earnings effect, both of which are important parameters for assessing the effectiveness of UI policy. If the effect of UI extensions on match quality is more positive in more concentrated labor markets, then this suggests that UI can bolster worker bargaining power and can serve as an additional policy tool beyond antitrust enforcement and the minimum wage for counteracting monopsony. If, instead, the UI-earnings effect is more negative in more concentrated labor markets, then policies intended to reduce high levels of concentration could have the additional benefit of improving the effectiveness of UI.

3 Policy Background and Data

In this section, I first discuss the policy context for my empirical strategy, followed by a description of the specific features of the LEHD data and the creation of my analysis sample.

3.1 UI Policy During the Great Recession

UI benefits in the U.S. typically last for 26 weeks, with some states offering more or less generous benefit durations. Additional UI benefits become available to qualifying unemployed workers during

periods of high unemployment in their state through two programs. First, the Extended Benefits (EB) program provides 13 or 20 weeks of benefits depending on the state's unemployment rate. EB is a permanent but optional program that is funded by both the federal and state governments, and not all states participate. During the Great Recession, however, the American Recovery and Reinvestment Act of 2009 temporarily changed EB to be fully federally funded, which induced several states to opt into the program (Rothstein, 2011).

The second program is the Emergency Unemployment Compensation program (EUC). Unlike EB, EUC is not permanent: Congress must authorize and fund it each time the need arises. During the Great Recession, EUC was initially authorized in June 2008 and provided up to 13 weeks of additional UI benefits for workers who exhausted their regular benefits at any point before the end of March 2009. As the recession deepened, Congress continued to re-authorize the program and expand the benefits offered. In November 2008, EUC was expanded to include two tiers: tier I provided eligible unemployed workers in all states with 20 weeks of benefits (7 beyond the original 13 weeks provided by the program), and tier II provided an additional 13 weeks for those in states with an unemployment rate above 6%. Another expansion in November 2009 changed tier II benefits to 14 weeks and created tiers III (13 weeks if the state unemployment rate exceeded 6%) and IV (6 weeks if the state unemployment rate exceeded 8.5%). At its most generous, EUC provided benefits for a total of 53 weeks beyond regular and EB benefits, for a potential total duration of 99 weeks.

This initial gradual expansion of benefits would be uncharacteristic of the remainder of EUC's existence, which Rothstein (2011) describes as "proceed[ing] in fits and starts." The program was reauthorized eight times after the November 2009 expansion, but also was allowed to expire three times between reauthorizations (in April, June, and November 2010). After each lapse, Congress eventually made benefits retroactive to the previous expiration of the program, though this feature was not always predictable: it took almost two months for retroactive benefits to be approved following the June 2010 expiration. In September 2012, EUC entered its final phase with benefits unchanged or less generous in tiers I through III and expanded in tier IV. The program ended in late December 2013.

This series of alterations, lapses, and reauthorizations of EUC motivates the assumption in my empirical strategy that nonemployed workers expect, at most, to receive the maximum potential

benefit duration available in their state at the time they separate from employment. Under program rules, higher tiers of benefits could only be accessed after qualifying for and exhausting benefits on all lower tiers. In addition, unemployed workers who exhausted benefits on lower tiers were immediately eligible for benefits on a higher tier as soon as their state triggered onto that tier. Beginning in 2010, the EUC program was consistently nearing expiration and even lapsed three times (Rothstein, 2011). Unemployed workers could not reasonably expect the continued availability of higher tiers of benefits.[8] Moreover, as the recovery progressed, states triggered off of higher EUC tiers as their labor markets gradually improved. Once a state no longer qualified for higher tiers of benefits, unemployed workers immediately lost access to those higher tier benefits. It is reasonable to conclude, therefore, that total potential UI benefits at the time of separation is the relevant policy parameter for the expectations formed by nonemployed workers about UI availability.

Eligible workers could claim benefits on a given EUC tier depending on their state's unemployment rate. EUC contained a series of triggers that determined which tier of benefits a state qualified for based on either of two measures of its unemployment rate. The first measure was the insured unemployment rate (IUR), which is the share of UI-covered workers in the state who are receiving regular UI benefits. The second measure is the total unemployment rate (TUR), which is the standard measure of the unemployment rate: the share of the labor force not working but who are ready to work and actively searching for a job. Each of these measures entered into the EUC trigger formulas as three-month moving averages to avoid high frequency movements between tiers.

Both the IUR and TUR were measured in real-time, and the Department of Labor issued weekly notices of both the latest readings of these unemployment rates and which states, if any, were eligible for each benefit tier (Chodorow-Reich et al., 2019). The IUR was measured using administrative data from the state's UI system and therefore was subject to minimal measurement error. The TUR, on the other hand, was measured using a combination of administrative and survey data. The Bureau of Labor Statistics (BLS) constructs state-level unemployment rates by combining estimated counts of employed and unemployed workers that are produced by a state-space filter. This state-space model uses as inputs the count of unemployed workers from the Current Population Survey (CPS), counts of insured unemployed workers, counts of employed workers from the CPS,

[8] Once EUC expired, UI beneficiaries could receive the remaining benefits on any tier that they had already started but could not access additional benefits on higher tiers.

and state-level payroll employment from the Current Employment Statistics (CES) program.[9]

The estimated state unemployment rates are subject to revision. As described in Chodorow-Reich et al. (2019), BLS revises the estimates because of: revisions to administrative data inputs (insured unemployed counts and CES employment), incorporation of the full time series of input data, and occasional methodological improvements. As discussed in Chodorow-Reich et al. (2019), from 1996 to 2015, the majority of cases in which state unemployment rates measured in real-time differed from the revised estimates were because of methodological changes to the BLS model. These differences between the real-time and revised data led to more than 600 instances, almost exclusively during the Great Recession, in which the maximum potential UI benefit duration for a given state and month would have been different if the revised unemployment rate had been used instead of the real-time unemployment rate.

The most relevant major methodological improvements for my sample period occurred in 2015, when BLS introduced better handling of state-specific outliers as well as improved seasonal adjustment. These updates are described by Bureau of Labor Statistics (2015) as leading to "more accurate and reliable estimates." Chodorow-Reich et al. (2019) estimate the importance of the various components of the revision process for the UI errors. They find that the 2015 methodological update and state-specific outlier detection explain about half of the variation in UI errors, while the incorporation of the full time series explains very little of the variation.

I use revised unemployment rates as of 2015 to construct my treatment variable, which is discussed in more detail in Section 4.1. Specifically, I follow Chodorow-Reich et al. (2019) and calculate UI errors, which are equal to the difference between potential UI durations using the real-time unemployment rate and counterfactual durations derived from the revised unemployment rate.[10] These UI errors isolate the cases when a state experienced a UI extension because of quasi-random measurement error and not a change in labor market fundamentals, under the assumption that the revised unemployment rate reflects true labor market conditions. As I discuss in more detail below, this measurement error approach allows me to invoke weaker identification assumptions than are typically required in the UI literature on benefit extensions in the U.S. context.

[9] Additional details about the estimation of state-level unemployment rates by BLS can be found in Chodorow-Reich et al. (2019) Online Appendix A.

[10] A natural assumption about the revision process is that UI errors are systematically correlated with state populations because they arise from survey-based measures with larger sample sizes in more populous states. Appendix Figure A.1 shows little correlation between state populations and UI errors.

3.2 Data

Understanding how the effects of UI extensions vary across the distribution of labor market concentration is a data-intensive exercise, requiring data linking workers to their employers in order to measure concentration, heterogeneity in local concentration, and variation in UI benefit duration. In the U.S. context, the primary source of variation in UI benefits is across states and over time. Therefore, I use data from the LEHD program at the U.S. Census Bureau, which is the most comprehensive linked employer-employee dataset for the U.S. that contains information from all states. The LEHD includes administrative data on employment and earnings reported by state UI agencies on a quarterly basis. A subset of states participates in any given LEHD project, and my sample consists of data from 28 states.[11] The data also include information on the age, geography, education, race, and ethnicity of workers. I also observe information about employers, such as industry, employment, and the number and location of establishments. In order to assign workers to a local labor market, which I define as the combination of a commuting zone and industry, I need to assign each worker to one industry. Appendix B describes how I assign workers to a single industry when multiple industries are associated with the same firm. In all cases, I use four-digit NAICS industries.

I collapse the LEHD panel into nonemployment spells defined as at least one quarter without earnings in any state.[12] For each spell, I also track quarterly earnings before and after the spell, the worker's employer and industry before and after the spell, and the state in which the worker was previously employed. In addition to this information on nonemployment spells, I add data on worker characteristics, commuting zone, and how long I observe them in the LEHD, which corresponds to actual labor market experience for many workers. I also include information on the worker's tenure at their last job prior to a nonemployment spell.

I do not observe UI eligibility or receipt. However, all workers in my sample are potentially eligible for UI at the time of hiring because all jobs included in the LEHD are covered by UI. The workers in my sample can lose UI eligibility if they voluntarily leave their job or fail to meet certain

[11] A subset of states participates in any given LEHD project. The states in my sample are: Alabama, Arizona, California, Colorado, Connecticut, Delaware, Idaho, Indiana, Kansas, Maine, Maryland, Nevada, New Jersey, New Mexico, New York, North Dakota, Ohio, Oklahoma, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, Washington, Wisconsin, and Wyoming.

[12] The LEHD includes an indicator for the presence of positive earnings in all states, regardless of whether they are in my sample.

"look back" criteria such as length of employment or total earnings, which vary by state. I make two sample restrictions in order to focus my analysis on workers likely to be eligible for UI at the time of separation. First, I only include prime-age workers (aged 25-54). This restriction reduces the possibility that a given nonemployment spell is actually the start of retirement for older workers or schooltime for younger workers. Second, I limit my attention to individuals who worked for at least two consecutive quarters at their prior job before the start of a nonemployment spell. I include this sample screen because most states require that unemployed workers have approximately six months of employment history in order to be eligible for UI benefits.[13]

For spells that are ongoing at the end of the sample window, I follow Nekoei and Weber (2017) and report estimates of the nonemployment duration effect for all spells as well as those that are completed within my sample window (96% of spells). This approach shows the importance of ongoing spells for my estimates of the duration elasticity. To reduce the influence of outliers in terms of the length of nonemployment spells, I top-code all nonemployment spells at 12 quarters (three years), which is a similar cap as employed by other papers in this literature (e.g. Nekoei and Weber, 2017). For estimates of the earnings elasticity, I only include spells where I observe at least two quarters of re-employment earnings.[14]

My analysis sample includes approximately 25,930,000 nonemployment spells experienced by 19,690,000 workers. Table 1 provides summary statistics for this sample. These workers are, on average, 39 years old and slightly more than half are women.[15] The workers are predominantly white (76%), with Black workers (14%) and Asian workers (6%) comprising the next largest groups. More than 80% of the sample holds a high school degree and 23% have at least a Bachelor's degree.[16]

[13]Further restricting the sample based on a minimum earnings requirement has little effect on sample composition or my estimates.

[14]I do not require that these two quarters of re-employment earnings be from full quarters of employment. I also do not require that they be at the same employer.

[15]The fact that slightly more than half of the nonemployment spells in my sample are experienced by women is at odds with the fact that men were hit harder by the Great Recession (Kochhar, 2011). This likely is a result of my inability to distinguish between nonemployment spells due to unemployment versus other causes, such as parental leave, which are much more likely to affect women. Appendix Table A.1 provides descriptive statistics for unemployed prime age workers from the Current Population Survey. Only 46% of these workers are women, and this proportion falls to 39% when looking at unemployed workers who are likely to be eligible for UI.

[16]The education information in the LEHD is a point-in-time snapshot and represents the highest level of education that the worker has completed at that time. Some nonemployed workers might enroll in further education in order to increase their human capital while searching for jobs, and this skill investment can translate into higher earnings once they are re-employed (Leung and Pei, 2020). Unfortunately, I do not observe the effects of further education, which potentially biases my results if there is differential re-enrollment along the distribution of UI errors and local labor market concentration.

Appendix Table A.1 provides demographic statistics for prime age unemployed workers from the Current Population Survey (CPS), including those likely to be eligible for UI. The workers in my sample have qualitatively similar characteristics as those in the CPS, with the exception of women, who are over-represented, and high school graduates, who are under-represented. On average, these workers are still relatively early in their careers, having about 12 years of total labor market experience. They are somewhat new to the jobs held prior to their nonemployment spells with an average tenure of just under 2.5 years.

The workers in my sample experienced an average of 2.6 nonemployment spells between 2004 and 2014. Because workers can separate from employment or start a new job at any point in a quarter, I measure nonemployment durations as the number of full quarters without any earnings. The average nonemployment spell in my sample is 2.2 quarters or 6.6 months, and the standard deviation is 1.9 quarters. The existence of partial quarters of employment at either end of a nonemployment spell also introduces noise into the comparison of earnings before and after nonemployment. For example, average earnings in the last quarter of employment before the start of a nonemployment spell is $\$6,378$, which is 22% lower than average earnings in the second-to-last quarter before a nonemployment spell ($\$8,205$). This difference could reflect partial quarters of employment, an Ashenfelter dip, or a combination of the two. A similar difference (25%) exists between average earnings in the first quarter of re-employment ($\$6,038$) and the second quarter of re-employment ($\$8,092$). In order to avoid issues related to partial quarters of employment at either end of a spell, my preferred outcome variable for earnings is the difference in log earnings between the second quarter after a nonemployment spell and the second-to-last quarter before a spell.

My empirical strategy uses variation in the UI error at the time of separation from employment. As described above, the UI error is measured in quarters and is the additional UI benefit duration available to nonemployed workers as a result of quasi-random measurement error. The average UI error experienced in my sample period is 0.0086 per quarter, which corresponds to an additional 0.1 weeks of potential UI benefits. This small average value is due to the quasi-random nature of the UI error and the fact that errors could be either positive, negative, or zero. There is still considerable variation in the UI error: its standard deviation in my sample is 0.1737. Figure 2 shows the distribution of UI errors during my sample period, both including (Panel A) and excluding zeros (Panel B). Focusing on the distribution of nonzero UI errors in Panel B, there is considerable mass

beyond 0.5 quarters (both positive and negative).

I measure labor market concentration using the Herfindahl-Hirschman Index (HHI) of quarterly payroll:

$$HHI_{z,j,t-1} = \sum_{k=1}^{K} s^2_{z,j,k,t-1}, \tag{1}$$

where $s_{z,j,k,t-1}$ is firm k's share of total payroll in the local labor market defined as the combination of commuting zone z and industry j in the quarter prior to the start of the nonemployment spell $t-1$. Payroll for firm k is equal to the number of workers at k times their earnings. I construct the payroll HHI using data from the Longitudinal Business Database (LBD). Unlike the LEHD, the LBD contains consistent firm identifiers across states, which allows me to accurately measure concentration within commuting zones that span multiple states. This version of the HHI was first used by Berger et al. (2022b). The average worker in my sample is exposed to an HHI of 0.14 in the quarter in which they separate from employment. This degree of concentration is just below the level of HHI (0.15) at which U.S. Department of Justice and Federal Trade Commission (2010) classifies markets as moderately concentrated. There is, however, considerable variation in concentration across local labor markets as the standard deviation is 0.21.

In addition to studying how UI extensions and labor market concentration influence earnings after nonemployment, I examine mechanisms related to the mobility of workers that potentially influence both the immediate and long-term effects of UI on earnings. In particular, I look at the extent to which longer UI benefits allow workers to switch workplaces and local labor markets. On average, 56% of nonemployment spells end with the worker switching SEINs, and 52% of spells end with the worker employed in a different local labor market. These switches across markets are more likely to be the result of changing industry but not commuting zone (48%) rather than moving to a different commuting zone (4%), though I cannot observe moves to states outside of my sample. Because concentration is associated with a variety of negative outcomes for workers, including lower earnings, this mobility is important because it might allow workers to match with an employer in a less concentrated market. Among the individuals that return to work within my sample window, the HHI in their local labor market of re-employment is 0.14, on average.

4 Empirical Strategy

The primary goal of my empirical strategy is to identify how local labor market concentration mediates the effect of UI extensions on labor market outcomes for nonemployed individuals. Before doing so, I follow the existing literature and first estimate the effect of UI extensions on match quality without taking into account local concentration.

4.1 UI Extensions and Match Quality

I begin by estimating models that relate changes in UI benefit duration to outcomes for nonemployed workers using models of the form:

$$Y_{itszj} = \delta T^*_{st} + \sum_{k=1}^{4} \pi_k UR_{s,t-k} + \theta X_{it} + \alpha_t + \rho_s + \gamma_{zj} + u_{itszj}, \tag{2}$$

where Y_{itszj} are labor market outcomes for worker i who began a nonemployment spell in quarter t after working in state s and in the local labor market defined by the combination of commuting zone z and industry j. The main outcomes I study are nonemployment duration and the change in earnings at re-employment, which is my preferred measure of match quality. The variable of interest is potential UI duration T^*_{st}: the maximum number of quarters of UI benefits for which a qualifying worker is potentially eligible after losing their job in quarter $t-1$ in state s.

I include fixed effects for the first quarter of the nonemployment spell (α_t) in order to compare outcomes for workers beginning a nonemployment spell at the same time. I also include state of employment fixed effects (ρ_s) to eliminate pre-existing differences across states in potential outcomes. These "spell start" and state fixed effects account for any correlation between outcomes, potential UI benefit duration, and labor market conditions that varies systematically over the course of the business cycle or across individual states.

Even after accounting for these factors, there still can be residual correlation between labor market outcomes and the length of UI benefits due to labor market conditions. For example, while UI policy is set at the state-level, local labor market conditions matter for how quickly workers return to work and their earnings once reemployed (e.g. Kroft et al., 2013). To control for systematic differences across local labor markets, I include commuting zone-by-industry fixed effects (γ_{zj}), which serve as local labor market fixed effects given how I define these markets. I also

include lags of the revised unemployment rate ($UR_{s,t-k}$), which allow me to compare outcomes for workers in states on similar labor market trajectories prior to a UI error. Worker-level controls (X_{it}) measured at the onset of a nonemployment spell are also included in order to absorb any remaining variation in the outcome variables. Specifically, I use fixed effects for age, education, gender, race and ethnicity, prior job tenure, and total observed labor market experience. Finally, u_{itszj} is the error term, and I cluster the standard errors at the state level because that is the unit at which UI policy varies.

This model identifies the effect of UI extensions on nonemployment duration and earnings under the assumption that, conditional on the controls, UI duration is not systematically correlated with workers' potential outcomes as they enter nonemployment. This assumption is unlikely to hold, even with the rich set of controls included in (2). As many researchers have noted, UI extensions occur when labor market conditions worsen. In the case of EUC and its triggers, the relationship between UI generosity and the unemployment rate is written into the law. The implication of this relationship is that the primary reason why UI durations vary across states is due to underlying differences in labor market conditions (Schmieder and von Wachter, 2016). The resulting mechanical correlation means that potential outcomes for nonemployed workers will be worse in states with extended benefits, and so the effects of UI identified off of cross-state variation can be biased by differences in underlying labor market conditions.

The endogenous relationship between potential benefit durations and labor market outcomes has motivated researchers to use empirical designs in which UI duration is orthogonal to labor market conditions. Almost all recent studies exploit discontinuities or kinks in the UI benefit schedule based on the age or experience of workers in several European countries (e.g. Card et al., 2007; Lalive, 2007; Schmieder et al., 2016; Nekoei and Weber, 2017). These research designs are generally not feasible in the U.S., where UI benefit duration is typically constant across all qualifying workers within a given state (Card et al., 2015; Johnston and Mas, 2018), and so researchers studying the U.S. context have relied on state-time variation in benefits instead (Rothstein, 2011; Farber and Valletta, 2015; Farber et al., 2015; Farooq et al., 2020).

In contrast, I use the measurement error approach, first developed by Chodorow-Reich et al. (2019), in order to overcome the endogeneity of UI extensions and labor market conditions. This approach allows me to control for actual labor demand and exploit plausibly exogenous variation

in potential UI benefit durations, conditional on that demand. By isolating differences in potential benefit durations that are due to measurement error and not underlying differences in labor market conditions, I can relax the assumption that UI duration and potential outcomes are conditionally uncorrelated.

To implement this strategy, I modify (2) to have the form:

$$Y_{itszj} = \beta_1 E_{st} + \sum_{k=1}^{4} \pi_k UR_{s,t-k} + \theta X_{it} + \alpha_t + \rho_s + \gamma_{zj} + \varepsilon_{itszj}. \tag{3}$$

The only difference between (2) and (3) is the treatment variable E_{st}, which is the UI error in the quarter of separation from employment, and the coefficient of interest β_1. The UI error is the difference in potential UI benefit duration when using the revised state-level unemployment rates and when using the real-time state-level unemployment rate. It is measured in quarters, and so the estimated coefficients from this model can be directly compared to those from (2). This specification approximates the experiment comparing outcomes for similar workers who separated from employment at the same time, worked in states with similar labor market and UI policy paths, but were quasi-randomly assigned to different UI benefit durations because of measurement error. The value of this design, as opposed to (2), is that the variation in potential benefit duration is purged of the component due to underlying labor market conditions, allowing for cleaner identification of the effect of UI extensions.

This model identifies the effect of UI extensions under the assumption that, conditional on the controls, the UI errors are exogenous with respect to the true underlying labor market conditions, which are accurately measured by the revised unemployment rate. Put another way, the UI errors are conditionally uncorrelated with potential outcomes for workers. In order to assess the validity of this identification assumption, I regress the UI error on both the contemporaneous and lagged state unemployment rates. The results of this test are presented in columns 1 and 2 of Table 2 and confirm that this assumption is likely to hold. In both cases, labor market conditions can explain at most 2.5% of the variation in the UI errors. As another test of the identification assumption, I regress the UI error on a set of worker characteristics. The results of this test are presented in columns 3 and 4 of Table 2 and indicate that the assumption is also likely to hold: the worker characteristics can explain at most 3.8% of the variation in the UI errors.

I also provide additional evidence that the UI errors are uncorrelated with potential outcomes for workers. In Table 3, I regress the UI error on predicted earnings while controlling for fixed effects for spell start, state, and local labor market of employment, as well as lags of the revised unemployment rate. Earnings are predicted in each of the two quarters prior to and following a nonemployment spell, using the same set of worker characteristics as in (3). This exercise provides a summary measure of residual selection conditional on the fixed effects and lagged unemployment rates. The results indicate no residual selection, as the coefficients, which are measured in dollars, are neither economically nor statistically significant: the coefficient with the largest magnitude is $29 for predicted earnings in the second-to-last quarter before a nonemployment spell.

As a final check on the validity of the identification assumption that UI errors are uncorrelated with potential outcomes for workers, I regress the UI errors on each of the worker characteristics included in (3) while controlling for the fixed effects and lags of the unemployment rate. The results are presented in Appendix Table A.2. The estimated coefficient for each characteristic is small, and it is only statistically significant for gender. I also regress the UI error on the total number of nonemployment spells that the worker experiences across the full sample, both before and after the focal UI error. This exercise can also be thought of as testing for selection in terms of otherwise unobservable worker quality. The coefficient from this regression is small and neither economically nor statistically significant. The evidence presented here strongly implies that the identification assumptions I invoke are likely to hold.

In a supplementary analysis, I estimate a similar regression to (3) in which the treatment variable is given by the actual UI benefit duration T^*_{st}, and I additionally include fixed effects for different levels of the revised unemployment rate. In the absence of measurement error, including potential benefit duration and the contemporaneous unemployment rate would lead to perfect collinearity and I could not estimate the model. The coefficient on potential benefit duration from this alternative approach is therefore identified solely off of measurement error. As discussed in Section 5, the estimates from this strategy are very similar to those from my preferred specification.

4.2 UI Extensions, Concentration, and Match Quality

To study how local labor market concentration influences the effect of UI extensions, I interact a measure of concentration with the UI error:

$$Y_{itszj} = \beta_1 E_{st} + \beta_2 HHI_{zjt} + \beta_3 E_{st} \times HHI_{zjt} + \sum_{k=1}^{4} \pi_k UR_{s,t-k} + \theta X_{it} + \alpha_t + \rho_s + \gamma_{zj} + \varepsilon_{itszj}. \quad (4)$$

Here, HHI_{zjt} is the payroll HHI measured in local labor market $z \times j$ in the quarter in which the worker separated from employment $(t-1)$. This model is identical to (3) except for the HHI_{zjt} and $E_{st} \times HHI_{zjt}$ terms. In addition to the identification assumption invoked in the previous subsection, the effect of UI extensions at different levels of local concentration is identified under the assumption that the effects of UI extensions in less concentrated local labor markets are an accurate counterfactual for the effects of UI extensions in more concentrated local labor markets. There are two potential threats to this identification assumption. First, there could be secular trends in outcomes around UI extensions that differ by concentration. Second, there could be unobserved shocks that correlate with the timing of UI extensions and that differ across the distribution of concentration within the same state.

These secular shocks to labor market conditions are unlikely to exist because they must be timed to align perfectly with the timing of UI errors, affect only a subset of local labor markets, and somehow be correlated with HHI. These requirements rule out any events or changes that affect an entire commuting zone or industry because the local labor markets within which I measure concentration are a combination of the two. Put differently, a commuting zone-wide shock would impact many local labor markets and would not necessarily be isolated to only high- or low-concentration markets. Similarly, an industry-wide shock would threaten identification only if it has differently sized effects in different commuting zones and those effects are correlated with concentration. In addition to these scenarios, which are unlikely to exist systematically, these shocks must also occur at the same time as the UI errors and be in the same direction. Given that overall labor market conditions explain very little of the variation in UI errors, it is also unlikely that any unobserved shocks correlate with the timing of these errors.

5 Results

In this section, I first discuss my estimates of the effect of UI extensions on nonemployment duration and earnings, and then demonstrate how local concentration mediates these effects.

5.1 The Overall Effect of UI Extensions

Before showing results from the measurement error approach, I present OLS estimates of the effect of UI extensions on nonemployment duration in Table 4. There are six columns that sequentially add controls in order to asses their importance for the estimates. Column 1 shows the correlation between nonemployment duration and potential UI duration, measured in quarters. Column 2 adds fixed effects for the quarter in which the nonemployment spell starts as well the state in which the worker was employed. This is my most basic model because these are the minimal controls necessary for my empirical approach. The point estimate is positive but not statistically significant at conventional levels. I next add four lags of the revised state unemployment rate in column 3. The point estimate doubles and is now significant at the 1% level, which highlights the importance of comparing workers in states on similar labor market trajectories. Column 4 introduces commuting zone-by-industry (i.e., local labor market) fixed effects, and the point estimate becomes slightly smaller but is qualitatively similar.

My preferred specification includes controls for worker characteristics and is shown in column 5. The point estimate is 0.0306, meaning that a one quarter (13 week) extension of UI benefits causes nonemployment spells to increase by 0.031 quarters or 0.4 weeks. This effect is statistically significant at the 1% level and represents an increase in nonemployment duration of 1.4% relative to the mean. The point estimate is very similar to that in column 4, meaning that controlling for worker characteristics contributes little additional explanatory power when estimating the effect of UI extensions on nonemployment durations. In order to assess the influence of spells that have not yet concluded (i.e., are right-censored) at the end of the sample window, I estimate duration effects among completed spells for which I observe at least one quarter of re-employment. This result is presented in column 6. The point estimate is barely changed, indicating that right-censored spells are not influencing my estimates.

Table 5 presents estimates using the measurement error approach. This table has the same

structure as Table 4. The point estimates in columns 2 and 3, which include spell start and state fixed effects and controls for lags of the revised unemployment rate, respectively, are nearly identical and show that additional UI benefits lengthen nonemployment duration. After including the local labor market fixed effects in column 4, the point estimate is approximately 20% smaller than in column 3 but tells a qualitatively similar story.

My preferred specification controls for worker characteristics and is shown in column 5. The point estimate, which is essentially unchanged from the specification without worker characteristics, indicates that extending UI benefits by one quarter causes nonemployment spells to be longer by 0.0629 quarters, or 0.8 weeks, which represents a 2.8% increase relative to the mean. This effect, while quite small, is statistically significantly different from zero at the 5% level. My ability to precisely estimate such small effects highlights the statistical power generated by my approach and data. In addition, the 95% confidence interval allow me to rule out duration effects longer than 1.4 weeks. This estimate is not driven by right-censored spells (column 6).

We would typically expect the measurement error-based estimates to be smaller than the OLS estimates because they should not suffer from residual endogeneity, but that is not the case here (Card and Levine, 2000). One reason for this pattern of results is that the UI errors capture the pooled effect of temporary extensions and reductions in UI generosity while the OLS estimates capture both temporary and permanent changes: the expiration of EUC in 2013 represented a large and permanent reduction in UI generosity that is not captured by the UI errors. Boone et al. (2021) note this difference between the policy changes under EUC and those at its expiration, and they also find pooled IV estimates that are larger than their OLS estimates.

To better understand how both extensions and reductions in UI duration affect outcomes, I compare positive (extensions) and negative (reductions) UI errors in Appendix Table A.3. The point estimates in column 1, which give the effect on nonemployment duration, show that positive UI errors cause nonemployment durations to be 0.01 quarters longer. The effect of negative UI errors is three-times larger in magnitude and negative: -0.04 quarters (0.5 week). This estimate is also much smaller than the findings in Johnston and Mas (2018), who estimate that a one-quarter decrease in UI benefit duration in Missouri caused nonemployment durations to decrease by 0.25 quarters (3.3 weeks).[17]

[17]The change in potential UI benefit duration studied by Johnston and Mas (2018) is actually a 16-week reduction in

Overall, my estimate of effect of UI extensions on nonemployment durations is smaller than most existing studies. Much of this literature is from the European context, where the majority of papers exploit UI policy that allows for age-based regression discontinuity (RD) studies in a handful of countries. As summarized in Schmieder and von Wachter (2016), estimates of the UI duration effect from Europe range from 0.05 to 0.65 (0.08-0.65 when only considering RD studies), implying that a one-quarter UI extension increases nonemployment spells by 0.7-8.5 weeks. My estimated effect of 0.8 weeks, therefore, is at the bottom end of this range.

In the U.S. context, most studies of UI extensions have focused on hazard probabilities and not marginal effects on nonemployment durations because of data limitations. Among the studies that have examined duration effects, their estimates range from 0.15-0.45 (2.0-5.9 weeks), with all recent studies estimating effects of at least 3.3 weeks (Schmieder and von Wachter, 2016). These recent studies are well identified as they use either RD/RK designs (Johnston and Mas, 2018; Landais, 2015) or a politically motivated change in benefits (Card and Levine, 2000). However, they are also limited to studying one or two states, meaning that it is unclear how well their effects generalize beyond the studied populations.[18]

I next investigate how these UI extensions impact the quality of new matches. My preferred measure of match quality is the difference in log earnings between the second quarter after a nonemployment spell and the second-to-last quarter before a nonemployment spell. This measure implicitly introduces a worker fixed effect into the regression, which controls for possible selection on unobservables, and the coefficients can be interpreted as approximately equal to the percent change in earnings. I present estimates using other earnings outcomes in Section 5.4.

Table 6 presents estimates of the effect of UI extensions on the earnings of workers who exit nonemployment using the measurement error approach. The corresponding OLS estimates are in Appendix Table A.4. Using the measurement error approach and my preferred specification in column 5, I find that extending UI benefits by one quarter causes earnings to increase by 0.9%. This estimate is not statistically significant at conventional levels, but is relatively precise as I can rule out negative effects beyond -1.1%. Despite its lack of significance, this point estimate is notable

benefits. They calculate an expected difference in average nonemployment duration between treatment and control groups of 1.1 weeks for a one-month reduction in benefits. I re-scale this one-month estimate to its one-quarter equivalent by multiplying their estimate by 3.

[18]Census Bureau rules require at least three states be included in any LEHD disclosure, preventing me from replicating these studies.

relative to the existing literature on match quality because only two other studies have estimated a positive effect of UI extensions on match quality. Nekoei and Weber (2017) estimate an increase in earnings of 0.5% following a nine-week extension in Austria, and Farooq et al. (2020) estimate a statistically significant 2.9% increase in quarterly earnings for one additional quarter of potential UI benefits using the same data and a longer time frame as this paper.[19]

This point estimate represents the effect for workers experiencing nonemployment spells of various lengths. If there is selection in terms of productivity across spells of different lengths, then this point estimate might mask different earnings effects at different nonemployment durations. To address earnings differences driven by this potential selection into employment, column 6 includes a fixed effect for the length of the nonemployment spell. This fixed effect is technically a "bad control" but allows me to diagnose the degree to which unobserved productivity differences drive my result. The effect size is essentially unchanged at 1.2% (not statistically significant), which supports the use of the log difference earnings measure to account for unobserved heterogeneity.

The results presented in this section indicate that the measurement error approach provides credible estimates of the effect of UI extensions. I next use this method to determine how the effects of UI vary across the distribution of local labor market concentration.

5.2 The Interaction of UI Extensions and Local Labor Market Concentration

Table 7 presents estimates of equation (4), which interacts the UI error with the HHI, both of which are measured in the quarter of separation. Column 1 shows estimates of the duration effect. The point estimate in the first row is the main effect of UI extensions and corresponds to the case of a perfectly competitive labor market in which the HHI is approaching zero. In such a market, a one quarter extension of UI benefits causes nonemployment durations to increase by 0.0617 quarters (0.80 weeks), which is very similar to the overall effect of UI extensions without accounting for concentration (0.0629 quarters in Table 5).

The coefficient on the interaction term, in row 2 of Table 7, is 0.0072, meaning that the effect of UI extensions on nonemployment durations is more positive at higher levels of concentration.

[19] Farooq et al. (2020) also report estimates for a more comparable time frame (2008-2013): a one-quarter extension leads to a 0.5% increase in quarterly earnings (not statistically significant). As discussed above, their approach compares workers exiting nonemployment at the same time and not entering nonemployment at the same time, and so their estimates are not directly comparable to mine.

The magnitude of this change, however, is neither economically nor statistically significant. For example, the effect of UI extensions when the HHI is 0.15, the upper bound for a market to be considered unconcentrated (U.S. Department of Justice and Federal Trade Commission, 2010), is 0.0628 quarters or 0.82 weeks. The effect when the HHI is 0.25, the lower bound for highly concentrated markets, is 0.0635 or 0.83 weeks. While this interaction effect is small, it is precisely estimated: for a difference in the HHI of 0.10, the difference between an unconcetrated and highly concentrated market, I can rule out effects larger than 0.16 weeks.

In contrast to the effect on duration, I find that the UI-earnings effect is statistically significantly different across the distribution of HHI. The main effect of UI extensions on earnings in the case of perfect competition (first row) is 1.7%, nearly twice as large as the overall effect on earnings without accounting for concentration (0.9% in Table 6). This point estimate is not statistically significant but is meaningfully large when compared to existing estimates of the UI-earnings effect discussed in the previous subsection. The interaction term is -0.0474 and statistically significant at the 5% level. This interaction term means the effect of UI on re-employment earnings is less positive at higher levels of local labor market concentration and eventually turns negative. For example, the UI-earnings effect is 1.0% in unconcentrated local labor markets (HHI = 0.15) and 0.5% in highly concentrated local labor markets (HHI = 0.25). In other words, the size of the UI-earnings effect is approximately 50% smaller in highly concentrated labor markets compared to unconcentrated labor markets. The interaction term also is precisely estimated, as I can rule out changes in the UI-earnings effect outside of the interval from -0.89% to -0.06% for a change in HHI of 0.10.

Taken together, the results in Table 7 indicate that the effect of UI extensions on nonemployment durations is constant across the distribution of local labor market concentration but that their effect on re-employment earnings declines at higher levels of concentration. This pattern of results implies that the smaller UI-earnings effect in more concentrated labor markets is not driven by longer nonemployment spells and negative duration dependence. Instead, my results suggest that the change in bargaining power provided by the UI extensions I study is insufficient to counteract the negative outcomes associated with higher levels of concentration. In terms of policy, these results imply that efforts to reduce high levels of local labor market concentration can additionally help the UI system facilitate better matches.

I also examine how UI affects outcomes at different parts of the distribution of concentration.

Figure 3 shows the estimated marginal UI-duration and UI-earnings effects across the distribution of local concentration. Consistent with the marginal effects shown in Table 7, the figure shows that UI-duration effect (Panel A) is essentially flat at all levels of HHI while the UI-earnings effect (Panel B) is less positive at higher levels of concentration and turns negative once the HHI exceeds 0.35. However, the confidence interval denoted by the shaded region implies that the marginal UI-earnings effect is not statistically significantly different from zero at any level of local concentration.

A potential concern with my preferred approach is the influence of extreme values of the HHI on my estimates. I prefer to use the level of HHI because the "outlying" values of the HHI distribution correspond to the cases with the least potential for employer market power (the perfectly competitive limit as the HHI approaches zero) and the greatest potential for such power (the monopsonistic limit as the HHI approaches one). But it is still important to understand the influence of observations at either end of the range of HHI. To further explore how extreme values of the HHI impact my results, Table A.6 shows estimates from a version of equation (4) using $\log(HHI)$. The coefficients in this table have a different interpretation as a result of the log transformation. The point estimates in the first row now correspond to the effect of UI extensions in the case of true monopsony, where $HHI = 1$. In such a market, a one quarter extension of UI benefits causes nonemployment durations to increase by 0.0766 quarters (1 week) and earnings at re-employment to remain unchanged (0.03%). The sign of the interaction term in row 2, for both duration and earnings, is the same as that using the level of HHI but the magnitude is smaller. At an HHI of 0.15, the marginal effect of an additional quarter of UI benefits is an increase in nonemployment duration of 3.0% relative to the mean duration and an increase in earnings at re-employment of 0.6%. These marginal effects are essentially unchanged at an HHI of 0.25: the UI-duration effect is 3.1% and the UI-earnings effect is 0.5%. These point estimates, together with the estimates in Table 7, imply that the significantly lower UI-earnings effect in more concentrated markets is driven by differences in the UI-earnings effect at the extremes of the HHI distribution. This pattern of effects also suggests that labor market interventions aimed at reducing the highest levels of concentration might improve the ability of UI to deliver better matches.

5.3 Mechanisms

The estimates in Section 5.2 show that the effect of UI extensions on earnings is more negative at higher levels of concentration. Are there ways other than the immediate impact on earnings through which UI extensions could affect match quality for these workers? To answer this question, I investigate the mobility of workers across the distribution of local concentration.

To the extent that concentration does translate into monopsony power, nonemployed workers exposed to higher levels of concentration might use extended UI benefits to switch employers or local labor markets to escape such power. Table 8 presents estimates of the effect of UI extensions on different measures of worker mobility. I find that the effect of UI extensions on the likelihood of changing workplaces and local labor markets increases at higher levels of concentration, but these effects are not economically significant. Among workers that do switch local labor markets after a nonemployment spell, there is no difference in the concentration of their new labor market compared to their old one.

5.4 Robustness

In this section, I explore the robustness of my results to an alternative empirical strategy and different earnings measures. My empirical strategy isolates the UI extensions caused by quasi-random errors in the real-time measurement of state-level labor market conditions. An alternative way to isolate the effect of this measurement error is to use as the treatment variable the actual potential UI durations based on the real-time unemployment rate while controlling for the concurrent real-time unemployment rate. In the absence of measurement error, these potential UI durations and unemployment rates would be perfectly collinear because UI duration is a direct function of the unemployment rate. The coefficient on potential UI benefit duration with this approach is therefore identified solely off of measurement error.

The results using this approach are given in Table 9. In terms of the duration effect, the coefficient in column 1 is 0.0434, which is between that from my OLS results and UI error results. This represents an increase in nonemployment duration of 1.9% relative to the mean, and is statistically significant at the 1% level. The estimate for earnings also is similar to my estimates using the UI errors: an additional quarter of UI benefits causes earnings to increase by 0.8%. It is reassuring that

I obtain similar estimates from this alternative approach because it emphasizes that the underlying identifying variation in my estimates using the UI error is coming from measurement error and not underlying labor market conditions.

I also examine the effect of UI extensions on alternative measures of earnings. These estimates are shown in Table A.5. In the first two columns, I estimate the effect of UI extensions on log earnings in the first quarter (column 1) and the second quarter (column 2) after a nonemployment spell, while controlling for log earnings two quarters prior to the start of the nonemployment spell. These estimates do not implicitly include a worker fixed effect and therefore include additional unobserved worker heterogeneity not present in my preferred measure of match quality. My results indicate that, without controlling for unobserved worker heterogeneity, UI extensions led to decreases in earnings of 1.4-2.3%. These effects, however, are imprecise and not statistically significant.

In column 3 of Table A.5, I use as the outcome variable the difference in log earnings between the first quarter of re-employment and the last quarter before a nonemployment spell. Like my preferred measure of match quality, this outcome implicitly controls for unobserved heterogeneity across workers. My results are very similar to my preferred specification: earnings increase by 0.8% following a UI extension, an effect that is not statistically significant. Taken together, the results using these alternative earnings measures point to the importance of accounting for unobserved heterogeneity and the robustness of my results to a related measure of match quality.

6 Conclusion

This paper provides estimates of the effect of UI extensions using a measurement error approach that is new to the study of UI and worker-level outcomes. I also show the first estimates of how additional UI benefits affect workers across the distribution of local labor market concentration. It is theoretically ambiguous how the effect of UI on match quality will vary across the distribution of concentration. UI extensions provide an additional outside option that can improve the bargaining power of workers exposed to higher levels of concentration, enabling them to be more selective in their search and leading to a more positive UI-earnings effect in more concentrated labor markets. Workers in highly concentrated local labor markets, however, might be limited in their ability to become more selective, and UI extensions could lead to longer nonemployment durations and a more

negative UI-earnings effect among these workers compared to those in less concentrated markets.

I identify the effects of UI extensions using measurement error in state unemployment rates that assigned conditionally random potential UI benefit durations during the Great Recession. Overall, I find that nonemployment durations are 2.8% longer and re-employment earnings are 0.9% higher following a UI extension, though only the duration estimate is statistically significantly different from zero. My estimated duration effect is comparable to recent estimates using regression discontinuity designs in Europe, but substantially smaller than U.S. studies based on policy changes in one or two states. My estimated earnings effect, meanwhile, is not significantly different from zero but nearly twice as large as the upper range of existing estimates. The sign and magnitude of this point estimate is notable given that the majority of the prior literature finds small and negative effects of UI extensions on earnings.

I also find that the impact of UI extensions on earnings is lower in more concentrated labor markets, an effect that is driven by differences at the extremes of the HHI distribution. As a result, the UI-earnings effect in unconcentrated labor markets is twice as large as that in highly concentrated labor markets, though neither effect is statistically significantly different from zero. The difference in the UI-earnings effect is not the result of UI extensions having larger effects on nonemployment durations in more concentrated labor markets. This pattern suggests that UI extensions during the Great Recession did not provide a sufficiently large increase in worker bargaining power to offset the negative outcomes associated with labor market concentration. In terms of mechanisms, I find limited evidence of differences in worker mobility across workplaces, labor markets, and industries across the distribution of concentration. UI extensions also do not induce workers to move to less concentrated labor markets.

My results have important implications for policy. In particular, they indicate that UI extensions during the Great Recession had a small impact on the time workers spent out of work, and that this additional time searching led to positive but not significant changes in their earnings. These findings imply very little scope for moral hazard as a result of longer UI benefits, consistent with prior research on the trade-off between moral hazard and liquidity at the heart of UI policy during recessions (e.g., Rothstein, 2011; Kroft and Notowidigdo, 2016).

In addition, by providing the first estimates of the interaction of UI and concentration, I help to inform policy related to labor market power, of which concentration is a potential source. My results

imply that UI extensions neither hurt nor help workers exposed to higher levels of concentration, which is important for policy makers concerned about unintentional negative effects of additional UI benefits during the next recession. Moreover, I find that the positive impact of UI extensions on earnings decreases at higher levels of concentration. If other policies, like antitrust enforcement, are able to reduce local labor market concentration, they might also have complementary benefits in the form of improving the effectiveness of UI.

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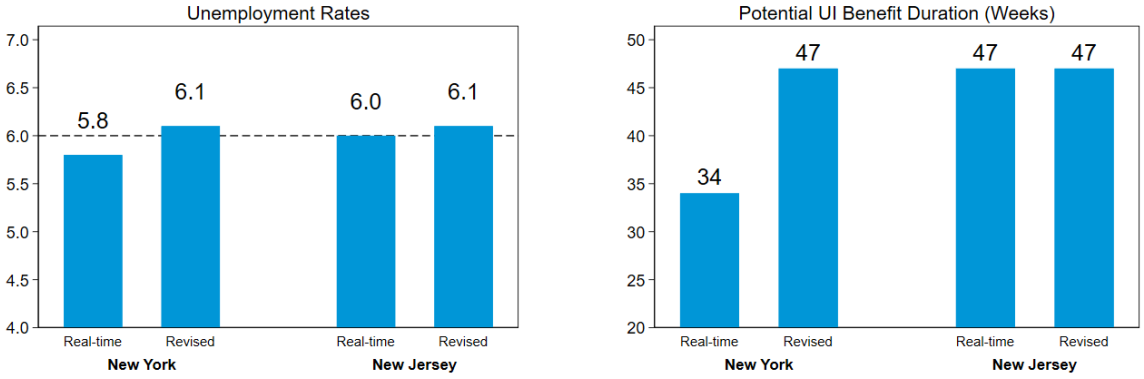
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Figures

Figure 1: Example of UI Error: January 2009

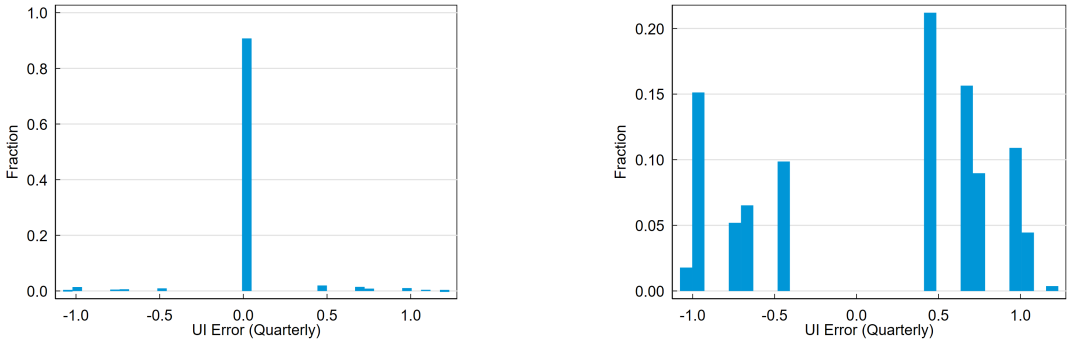


(a) Unemployment rates

(b) Potential benefit duration (weeks)

This figure provides an example of the UI error, which is calculated as the difference in potential UI benefit duration under the real-time state-level unemployment rate and the counterfactual benefit duration using the revised state-level unemployment rate. The Emergency Unemployment Compensation Program (EUC) from 2008 provided an additional 13 weeks of benefits in states with an unemployment rate of at least 6.0%. In this example, using the real-time unemployment rates, UI-eligible workers in New York were eligible for 34 weeks of UI benefits, while those in New Jersey were eligible for 47 weeks of benefits. Using the revised unemployment rates, workers in both states would have been eligible for 47 weeks of benefits. The UI error for New York is, therefore, $34 - 47 = -13$ weeks. In New Jersey, it is $47 - 47 = 0$ weeks.

Figure 2: Distribution of UI Errors

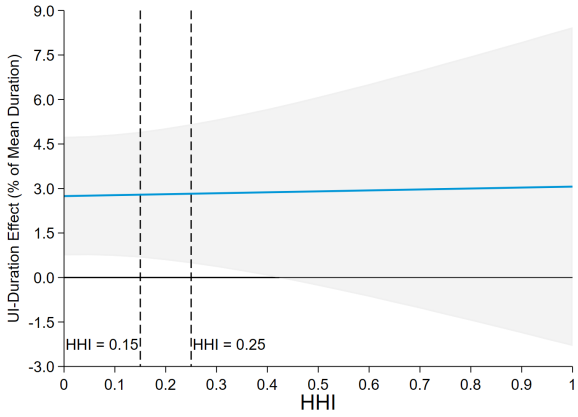


(a) Quarterly UI Error

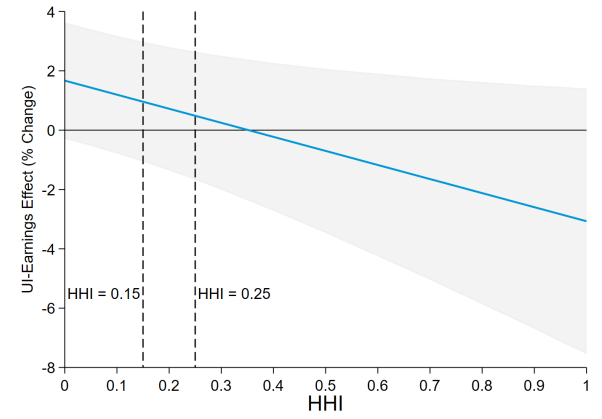
(b) Nonzero Quarterly UI Errors

Histograms of the size of UI Errors in the 28 states in my sample from 2004-2014. Panel A includes all state-quarter observations. Panel B includes all state-quarter observations with nonzero UI Errors. UI Error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter (measured in quarters).

Figure 3: Marginal Effects of UI Extensions



(a) Nonemployment Duration



(b) Earnings

These figures plot the marginal effects of UI extensions on nonemployment duration and earnings based on the estimates in Table 7 at different values of HHI. Shaded areas denote 95% confidence intervals. Dashed vertical lines indicate cutoffs for low concentration (HHI = 0.15) and high concentration (HHI = 0.25) markets according to U.S. *Horizontal Merger Guidelines* (U.S. Department of Justice and Federal Trade Commission, 2010).

Tables

Table 1: Summary Statistics

	(1) Mean	(2) SD
Age	38.78	8.872
Women	0.5282	0.4992
White	0.7626	0.4255
Black	0.1405	0.3475
Asian	0.0600	0.2375
Other races	0.0369	0.1885
Hispanic	0.2060	0.4044
Less than high school	0.1852	0.3884
High school	0.2786	0.4483
Some college/Associates degree	0.3092	0.4622
Bachelors degree or higher	0.2270	0.4189
Earnings in final quarter before nonemployment spell[a]	6,378	8,551
Earnings in second-to-last quarter before nonemployment spell[a]	8,205	8,414
Earnings in first quarter after nonemployment spell[a]	6,038	8,003
Earnings in second quarter after nonemployment spell[a]	8,092	8,155
Total labor market experience[b]	49.38	20.86
Prior job tenure[b]	9.555	11.39
Number of nonemployment spells	2.628	1.801
Nonemployment spell length[b]	2.249	1.921
UI error	0.0086	0.1737
Payroll HHI in quarter of separation	0.1383	0.2078
Payroll HHI at re-employment	0.1368	0.2064
Total Payroll in Local Labor Market[a]	47,000	247,000
Different SEIN	0.5641	0.4959
Different local labor market	0.5194	0.4996
Different industry (same commuting zone)	0.4806	0.4996
Different commuting zone	0.0388	0.1930
Observe at least 1 quarter of re-employment	0.9551	0.2072
Observe at least 2 quarters of re-employment	0.7563	0.4293
Number of nonemployment spells	25,930,000	
Number of workers	19,690,000	

a) 2018$, winsorized at the 1% and 99% levels. b) Measured in full quarters of nonemployment (top-coded at 12). Sample sizes and estimated values have been rounded for disclosure avoidance.

Table 2: Selection: Exogeneity of UI Errors and Labor Market Conditions

	(1)	(2)	(3)	(4)
R^2	0.0004	0.0242	0.0144	0.0384
Concurrent revised unemployment rate	X			
Lagged revised unemployment rate		X		X
Worker characteristics			X	X
Observations	25,930,000	25,930,000	25,930,000	25,930,000

Each column is a separate regression. In column 1, the UI error is regressed on the revised unemployment rate in the quarter of separation from employment. In column 2, the UI error is regressed on four lags of the revised unemployment rate. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance.

Table 3: Selection: UI Errors and Predicted Earnings

	(1) 2 Quarters Before Spell	(2) 1 Quarter Before Spell	(3) 1 Quarter After Spell	(4) 2 Quarters After Spell
UI error	28.51	12.59	-11.05	-5.93
	(54.75)	(43.53)	(33.85)	(38.34)
Spell start FE	X	X	X	X
State FE	X	X	X	X
Local labor market FE	X	X	X	X
Lagged revised unemployment rate	X	X	X	X
Observations	25,930,000	25,930,000	20,370,000	16,700,000
R^2	0.1393	0.1385	0.1462	0.1506

Each column is a separate regression. The outcome variable is predicted quarterly earnings in the quarter indicated in the column title. Earnings are predicted using a model consisting of fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. The coefficients are measured in dollars Sample sizes and estimated values have been rounded for disclosure avoidance.

Table 4: Nonemployment Duration – OLS Estimates

	(1)	(2)	(3)	(4)	(5)
Potential UI Duration	0.0162	0.0361***	0.0313***	0.0306***	0.0304***
	(0.0137)	(0.0108)	(0.0091)	(0.0090)	(0.0091)
% Effect	0.72	1.61	1.39	1.36	1.49
Spell start FE	X	X	X	X	X
State FE	X	X	X	X	X
Local labor market FE			X	X	X
Lagged unemployment rate		X	X	X	X
Worker characteristics				X	X
Spell length FE					
Completed Spells Only					X
Observations	25,930,000	25,930,000	25,930,000	25,930,000	24,760,000
R^2	0.5389	0.5389	0.5506	0.5542	0.1213
Dep. Var. Mean	2.249	2.249	2.249	2.249	2.037

In each column, the dependent variable is the length of nonemployment spells measured in quarters (top-coded at 12). Potential UI Duration is the maximum number of quarters of UI benefits available to qualifying workers who separate from a job in the quarter before the onset of a nonemployment spell. Local labor market fixed effects are commuting zone-by-industry (4-digit NAICS level) fixed effects. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

Table 5: Nonemployment Duration – Measurement Error Approach

	(1)	(2)	(3)	(4)	(5)
UI error	0.0825***	0.0821**	0.0639**	0.0629**	0.0641**
	(0.0293)	(0.0311)	(0.0254)	(0.0243)	(0.0239)
% Effect	3.67	3.65	2.84	2.80	3.15
Spell start FE	X	X	X	X	X
State FE	X	X	X	X	X
Local labor market FE			X	X	X
Lagged unemployment rate		X	X	X	X
Worker characteristics				X	X
Spell length FE					
Completed Spells Only					X
Observations	25,930,000	25,930,000	25,930,000	25,930,000	24,760,000
R^2	0.5389	0.5390	0.5506	0.5542	0.1214
Dep. Var. Mean	2.249	2.249	2.249	2.249	2.037

In each column, the dependent variable is the length of nonemployment spells measured in quarters (top-coded at 12). UI error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter before the nonemployment spell onset (measured in quarters). Local labor market fixed effects are commuting zone-by-industry (4-digit NAICS level) fixed effects. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

Table 6: Re-employment Earnings – Measurement Error Approach

	(1)	(2)	(3)	(4)	(5)
UI error	-0.0091	-0.0059	0.0004	0.0093	0.0122
	(0.0084)	(0.0080)	(0.0077)	(0.0106)	(0.0099)
% Effect	-0.91	-0.59	0.04	0.93	1.22
Spell start FE	X	X	X	X	X
State FE	X	X	X	X	X
Local labor market FE			X	X	X
Lagged unemployment rate		X	X	X	X
Worker characteristics				X	X
Spell length FE					X
Completed Spells Only	X	X	X	X	X
Observations	16,700,000	16,700,000	16,700,000	16,700,000	16,700,000
R^2	0.0019	0.0020	0.0235	0.0816	0.0834
Dep. Var. Mean	-0.1213	-0.1213	-0.1213	-0.1213	-0.1213

In each column, the dependent variable is the difference in log earnings between the second quarter after a nonemployment spell and the second-to-last quarter before a nonemployment spell. UI error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter before the nonemployment spell onset (measured in quarters). Local labor market fixed effects are commuting zone-by-industry (4-digit NAICS level) fixed effects. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

Table 7: Interaction Effects of UI Extensions and Local Concentration

	(1) Nonemployment Duration	(2) Re-employment Earnings
UI error	0.0617**	0.0167
	(0.0229)	(0.0100)
UI error × HHI	0.0072	-0.0474**
	(0.0576)	(0.0211)
Marginal Effect at DOJ Thresholds		
Low Concentration Market (HHI = 0.15)	2.8%	0.96%
High Concentration Market (HHI = 0.25)	2.8%	0.49%
Spell start FE	X	X
State FE	X	X
Local labor market FE	X	X
Lagged unemployment rate	X	X
Worker characteristics	X	X
Spell length FE		
Completed Spells Only		X
Observations	25,930,000	16,700,000
R^2	0.5543	0.0816
Dep. Var. Mean	2.249	-0.1213

Each column is a separate regression. In column 1, the dependent variable is the length of nonemployment spells measured in quarters (top-coded at 12). In column 2, the dependent variable is the difference in log earnings between the second quarter after a nonemployment spell and the second-to-last quarter before a nonemployment spell. UI error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter before the nonemployment spell onset (measured in quarters). Department of Justice (DOJ) thresholds correspond to the upper bound for an unconcentrated market and the lower bound for a highly concentrated market according to U.S. Department of Justice and Federal Trade Commission (2010). Local labor market fixed effects are commuting zone-by-industry (4-digit NAICS level) fixed effects. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

Table 8: Interaction Effects of UI Extensions and Local Concentration: Mobility

	(1)	(2)	(3)	(4)	(5)
	Different SEIN	Different LLM	Different Industry	Different Commuting Zone	Next HHI
UI error	0.0059	0.0053	0.0016	0.0038*	-0.0024
	(0.0098)	(0.0100)	(0.0104)	(0.0022)	(0.0015)
UI error × HHI	0.0386*	0.0403*	0.0501***	-0.0098	-0.0015
	(0.0219)	(0.0236)	(0.0170)	(0.0093)	(0.0024)
% Effect: ΔHHI = 0.10	0.68	0.78	1.04	-2.53	-0.12
Spell start FE	X	X	X	X	X
State FE	X	X	X	X	X
Local labor market FE	X	X	X	X	X
Lagged unemployment rate	X	X	X	X	X
Worker characteristics	X	X	X	X	X
Spell length FE					
Completed Spells Only	X	X	X	X	X
Changed LLMs					X
Observations	20,370,000	20,370,000	20,370,000	20,370,000	6,000,000
R^2	0.1895	0.1707	0.1495	0.0456	0.1866
Dep. Var. Mean	0.5641	0.5194	0.4806	0.0388	0.1294

Each column is a separate regression. In column 1, the dependent variable is an indicator for changing SEINs following a nonemployment spell. SEINs can correspond to individual or multiple establishments. In column 2, the dependent variable is an indicator for changing local labor markets (LLMs), either as a result of changing industry or commuting zone or both. In column 3, the dependent variable is an indicator for changing industry but not commuting zone. In column 4, the dependent variable is an indicator for changing commuting zones. In column 5, the dependent variable is the HHI for the local labor market of re-employment in the first quarter after a nonemployment spell. The sample in column 5 is limited to spells in which the worker changed local labor markets. UI error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter before the nonemployment spell onset (measured in quarters). Local labor market fixed effects are commuting zone-by-industry (4-digit NAICS level) fixed effects. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

Table 9: Alternative Specification – Measurement Error Approach

	(1) Duration	(2) Earnings
UI error	0.0434***	0.0075
	(0.0149)	(0.0075)
% Effect	1.93	0.75
Spell start FE	X	X
State FE	X	X
Local labor market FE	X	X
Lagged unemployment rate	X	X
Worker characteristics	X	X
Spell length FE		
Completed Spells Only		X
Revised unemployment rate FE	X	X
Observations	25,930,000	16,700,000
R^2	0.5556	0.0822
Dep. Var. Mean	2.249	-0.1213

Each column is a separate regression. In column 1, the dependent variable is the length of nonemployment spells measured in quarters (top-coded at 12). In column 2, the dependent variable is the difference in log earnings between the second quarter after a nonemployment spell and the second-to-last quarter before a nonemployment spell. Potential UI Duration is the maximum number of quarters of UI benefits available to qualifying workers who separate from a job in the quarter before the onset of a nonemployment spell. Local labor market fixed effects are commuting zone-by-industry (4-digit NAICS level) fixed effects. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Revised unemployment rate fixed effect corresponds to the revised unemployment rate in the quarter of separation. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

A Additional Results

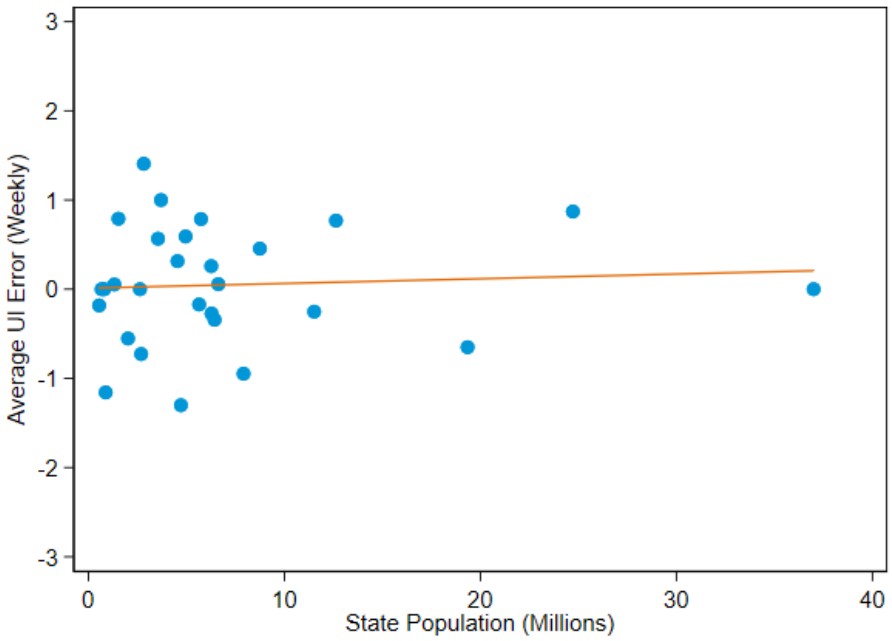


Figure A.1: UI Errors and State Populations

Scatter plot of the average weekly UI error and average state populations (in millions) for the 28 states in the sample from the period 2004-2014. UI error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter before the nonemployment spell onset (measured in weeks).

Table A.1: Summary Statistics - Current Population Survey

	(1) All Unemployed	(2) Unemployed and Likely UI Eligible
Age	38.7574	39.6563
	(8.8032)	(8.6946)
Women	0.4637	0.3912
	(0.4987)	(0.4880)
White	0.7377	0.7613
	(0.4399)	(0.4263)
Black	0.1785	0.1660
	(0.3829)	(0.3721)
Asian	0.0369	0.0318
	(0.1885)	(0.1755)
Other races	0.0469	0.0409
	(0.2115)	(0.1980)
Hispanic	0.1664	0.1693
	(0.3725)	(0.3750)
Less than high school	0.1670	0.1639
	(0.3730)	(0.3701)
High School	0.3622	0.3822
	(0.4806)	(0.4859)
Some college/Associates degree	0.2828	0.2743
	(0.4504)	(0.4462)
Bachelors degree or higher	0.1880	0.1796
	(0.3907)	(0.3839)
Unemployment spell length[a]	2.1587	2.0500
	(2.4274)	(2.3191)
Observations	329,288	214,570

a) Measured in quarters. Data from the January 2004-December 2014 waves of the Current Population Survey. Likely UI eligible means respondent indicated that they are unemployed as a result of job loss. Sample limited to prime age workers (aged 25-54).

Table A.2: Selection: UI Errors and Worker Characteristics

	(1) Age	(2) Women	(3) White	(4) Black	(5) Asian
UI error	0.0019	0.0039**	0.0019	-0.0019	0.0005
	(0.0354)	(0.0017)	(0.0017)	(0.0017)	(0.0005)
	(6) Other Races	(7) Hispanic	(8) < High School	(9) High School	(10) Some College
UI error	-0.0005	-0.0015	0.0010	-0.0002	-0.0010
	(0.0004)	(0.0016)	(0.0009)	(0.0006)	(0.0008)
	(11) College	(12) LM Experience	(13) Tenure	(14) # Spells	
UI error	0.0001	-0.1044	0.0902	-0.0213	
	(0.0.0007)	(0.2430)	(0.1640)	(0.0201)	

Observations = 25,930,000. All models control for lagged unemployment rate and fixed effects for spell start, state, and local labor market.

Table A.3: Positive and Negative UI Errors

	(1) Spell Length	(2) Earnings
Positive UI error	0.0103	0.0006
	(0.0142)	(0.0049)
Negative UI error	-0.0357*	-0.0088
	(0.0190)	(0.0079)
% Effect: Positive	0.46	0.06
% Effect: Negative	-1.59	-0.88
Spell start FE	X	X
State FE	X	X
Local labor market FE	X	X
Lagged unemployment rate	X	X
Worker characteristics	X	X
Spell length FE		
Completed Spells Only		X
Observations	25,930,000	16,700,000
R^2	0.5542	0.0816
Dep. Var. Mean	2.249	-0.1213

Each column is a separate regression. In column 1, the dependent variable is the length of nonemployment spells measured in quarters (top-coded at 12). In column 2, the dependent variable is the difference in log earnings between the second quarter after a nonemployment spell and the second-to-last quarter before a nonemployment spell. UI error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter before the nonemployment spell onset (measured in quarters). Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

Table A.4: Re-employment Earnings – OLS Estimates

	(1)	(2)	(3)	(4)	(5)
Potential UI Duration	0.0082*	0.0052	0.0071*	0.0098*	0.0111**
	(0.0047)	(0.0046)	(0.0040)	(0.0053)	(0.0053)
% Effect	0.82	0.52	0.71	0.98	1.11
Spell start FE	X	X	X	X	X
State FE	X	X	X	X	X
Local labor market FE			X	X	X
Lagged unemployment rate		X	X	X	X
Worker characteristics				X	X
Spell length FE					X
Completed Spells Only	X	X	X	X	X
Observations	16,700,000	16,700,000	16,700,000	16,700,000	16,700,000
R^2	0.0019	0.0020	0.0235	0.0816	0.0834
Dep. Var. Mean	-0.1213	-0.1213	-0.1213	-0.1213	-0.1213

Each column is a separate regression. In each column, the dependent variable is the difference in log earnings between the second quarter after a nonemployment spell and the second-to-last quarter before a nonemployment spell. Potential UI Duration is the maximum number of quarters of UI benefits available to qualifying workers who separate from a job in the quarter before the onset of a nonemployment spell. Local labor market fixed effects are commuting zone-by-industry (4-digit NAICS level) fixed effects. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

Table A.5: Alternative re-employment Earnings – Measurement Error Approach

	(1) Log Earnings: 1st Quarter	(2) Log Earnings: 2nd Quarter	(3) Log Difference: 1st Quarter
UI error	-0.0226	-0.0139	0.0075
	(0.0226)	(0.0109)	(0.0137)
% Effect	-2.26	-1.39	0.75
Spell start FE	X	X	X
State FE	X	X	X
Local labor market FE	X	X	X
Lagged unemployment rate	X	X	X
Worker characteristics	X	X	X
Spell length FE			
Completed spells only	X	X	X
Earnings 2 quarters before spell start	X	X	X
Observations	16,700,000	16,700,000	16,700,000
R^2	0.3056	0.2993	0.0299
Dep. Var. Mean	8.034	8.425	-0.0606

Each column is a separate regression. In column 1, the dependent variable is log earnings the last quarter before the onset of a nonemployment spell. In column 2, the dependent variable is log earnings the second-to-last quarter before the onset of a nonemployment spell. In column 3, the dependent variable is the difference between log earnings in the first quarter after a nonemployment spell and the first quarter before a nonemployment spell. UI error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter before the nonemployment spell onset (measured in quarters). Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

Table A.6: Interaction Effects of UI Extensions and Log Local Concentration

	(1) Nonemployment Duration	(2) Re-employment Earnings
UI error	0.0766*	0.0003
	(0.0430)	(0.0172)
UI error × Log(HHI)	0.0048	-0.0031
	(0.0079)	(0.0035)
Marginal Effect at DOJ Thresholds		
Low Concentration (HHI = 0.15)	3.0%	0.6%
High Concentration (HHI = 0.25)	3.1%	0.5%
Spell start FE	X	X
State FE	X	X
Local labor market FE	X	X
Lagged unemployment rate	X	X
Worker characteristics	X	X
Spell length FE		
Completed Spells Only		X
Observations	25,930,000	16,700,000
R^2	0.5543	0.0816
Dep. Var. Mean	2.249	-0.1213

Each column is a separate regression. In column 1, the dependent variable is the length of nonemployment spells measured in quarters (top-coded at 12). In column 2, the dependent variable is the difference in log earnings between the second quarter after a nonemployment spell and the second-to-last quarter before a nonemployment spell. UI error is the difference in potential UI benefit duration using the real-time and revised state unemployment rate in the quarter before the nonemployment spell onset (measured in quarters). Department of Justice (DOJ) thresholds correspond to the upper bound for an unconcentrated market and the lower bound for a highly concentrated market according to U.S. Department of Justice and Federal Trade Commission (2010). Local labor market fixed effects are commuting zone-by-industry (4-digit NAICS level) fixed effects. Worker characteristics include fixed effects for gender, education, race and ethnicity, age, total labor market experience, and prior job tenure. Sample sizes and estimated values have been rounded for disclosure avoidance. Standard errors clustered at the state level in parentheses: significant at *10%, **5%, and ***1%.

B Data Appendix

The variable identifying employers in the LEHD is the State Employer Identification Number (SEIN). SEINs do not necessarily correspond to establishments for firms with multiple locations, and multiple industries can be associated with the same SEIN if it happens to have multiple establishments (SEINUNITs). In order to assign workers to a local labor market, which I define as the combination of a commuting zone and industry, I need to assign each worker to one industry. This assignment presents two challenges. First, workers may hold multiple jobs. For these workers, I assign them to a "dominant employer" in each quarter. I follow Hyatt et al. (2014) and define the dominant employer as that with the highest combined earnings in two consecutive quarters. Second, firms with multiple establishments might have one SEIN, and I do not observe the actual SEINUNIT for workers at firms that happen to report separately by establishment. I address this challenge by assigning workers employed by a SEIN associated with multiple industries to the one with the most employment in their county of residence.